

RISHAB VEMPATI

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EDUCATION

Purdue University

B.S. in Computer Science & Artificial Intelligence, GPA: 3.83/4.00

West Lafayette, IN

Aug. 2023 – May 2027

EXPERIENCE

Software Engineering Intern

Aug. 2025 – Present

Ekai (Startup)

Remote

- Built Microsoft Teams integration for enterprise AI assistant, adapting existing Slack implementation to support Teams-specific features including automated task execution and knowledge base querying.
- Integrated Microsoft Graph API with RAG pipeline to enable document retrieval across 10,000+ files using Pinecone vector database, achieving 92% retrieval accuracy.
- Optimized semantic search with hybrid SQL retrieval and caching layer, reducing P95 response latency from 8.2s to 1.8s for 50+ concurrent users.

Software Engineering Intern

Jun. 2025 – Aug. 2025

AssetMark

San Francisco, CA

- Designed and implemented migration of legacy Enterprise Communication Service from monolithic .NET/C# to event-driven microservices using Azure Service Bus, processing 1M+ emails quarterly with 99.7% delivery success rate.
- Designed fault-tolerant message handling with dead-letter queues and exponential backoff retry policies, reducing message failure rate from 0.8% to 0.12% (85% reduction) and eliminating manual intervention for errors.
- Decoupled 5 service dependencies into independently deployable microservices with API gateways, enabling horizontal autoscaling that handled traffic spikes during quarter-end reporting periods.

Software Engineering Intern

May 2024 – Aug. 2024

Accelera Payments (Startup)

San Francisco, CA

- Implemented Kafka streaming pipeline for ISO 20022 payment processing with partition-based load balancing and exactly-once delivery semantics, load-tested to 50K+ daily transactions.
- Containerized 4 microservices using Docker with multi-stage builds in GitLab CI/CD pipelines, reducing deployment time from 2 hours to 25 minutes and enabling zero-downtime blue-green releases.
- Built integration test suite and Grafana dashboards monitoring message throughput (avg 850 msg/sec), processing latency (P99: 120ms), and error rates across distributed system.

Undergraduate Researcher – Data Science

Jan. 2024 – May 2024

Cisco

West Lafayette, IN

- Constructed demand forecasting models using ensemble methods (Random Forest, XGBoost) with feature engineering on 2M+ records, improving MAPE by 18% over baseline ARIMA models across 10,000 product SKUs.
- Optimized ETL pipeline using Apache Spark for distributed processing of supply chain datasets, implementing incremental loading that reduced execution time from 45 minutes to 18 minutes.

PROJECTS

Parking Lot Detection & Mapping | Python, PyTorch, OpenCV, Docker, AWS |

- Collaborated on 5-person team developing U-Net CNN in PyTorch for satellite image segmentation, achieving 84% IoU accuracy on parking lot detection across 10K+ validation images.
- Built automated pipeline ingesting 50GB+ Bing Maps imagery; containerized training workflow with Docker, reducing training time by 60%; contributed 5,000+ annotations to OpenStreetMap.

YourPingMe – Intelligent Notification Scheduler | Python, React, Redis, Celery, PostgreSQL |

- Designed task automation platform using Redis and Celery for scheduling recurring LLM API calls with cron expressions; built notification system delivering real-time SMS/email alerts upon task completion.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, SQL, JavaScript, TypeScript, Bash, R, HTML/CSS, Assembly

Technologies: FastAPI, Flask, Node.js, .NET, Docker, Kubernetes, Apache Kafka, Redis, CI/CD, Linux, Azure, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB

AI/ML: PyTorch, TensorFlow, JAX, OpenCV, Pinecone, LangChain, Elasticsearch, LangGraph, MCP, Spark

Certifications: AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals (AZ-900)